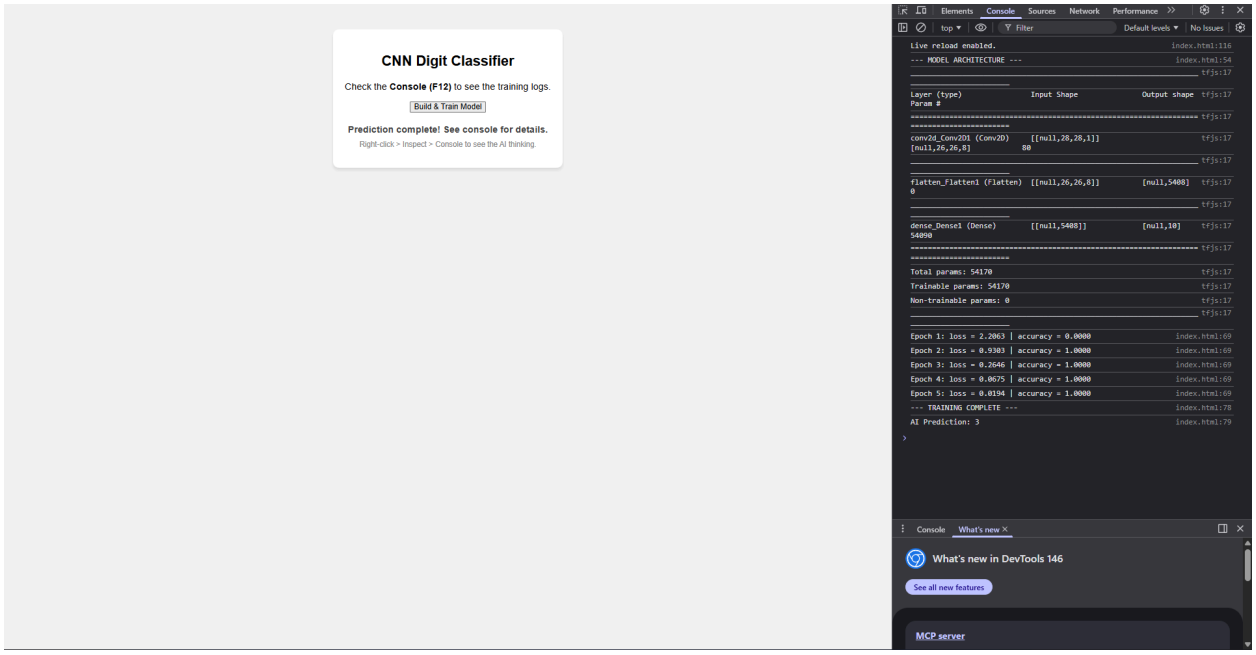
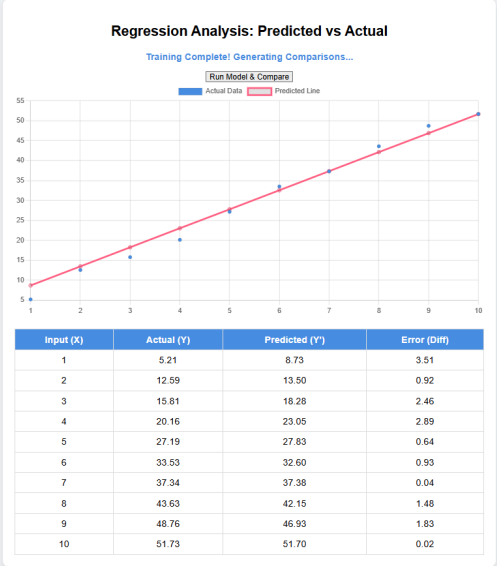




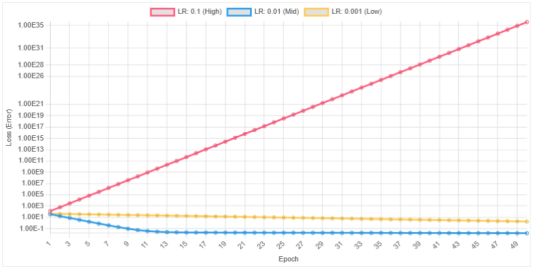
Input (X)	Actual (Y)	Predicted (Y')	Error (Diff)
1	5.21	8.73	3.51
2	12.59	13.50	0.92
3	15.81	18.28	2.46
4	20.16	23.05	2.89
5	27.19	27.83	0.64
6	33.53	32.60	0.93
7	37.34	37.38	0.04
8	43.63	42.15	1.48
9	48.76	46.93	1.83
10	51.73	51.70	0.02

Experiment: Learning Rate & Convergence

Watch how the "Loss" (error) drops differently for each learning rate.

Start Experiment

Fast (0.1) Medium (0.01) Slow (0.001)



Generalization Test: Predicting Unseen Data

We train the AI on inputs 1-10, then ask it to predict values for 15, 20, and 25.

Training Data (Seen):
Inputs: [1, 2, 3, ..., 10]
Rule: y = 10x - 5

Test Data (Unseen):
Inputs: **[15, 20, 25]**

Train & Predict Unseen

AI Performance on Unseen Data:

Input: 15
Expected: 145 | AI Predicted: 131.10
Accuracy Gap: 13.90

Input: 20
Expected: 195 | AI Predicted: 172.68
Accuracy Gap: 22.32

Input: 25
Expected: 245 | AI Predicted: 214.26
Accuracy Gap: 30.74